Lesson 02 Demo 01

Negotiating Response Format for Multi-Agent Communication

Objective: To demonstrate how multi-agent systems maintain reliable communication across mismatched data formats by declaring capabilities, negotiating a shared format at runtime, and automatically converting payloads (for example, XML ↔ JSON) when needed, while failing safely with clear, auditable logs when no common format exists

Tools required: None

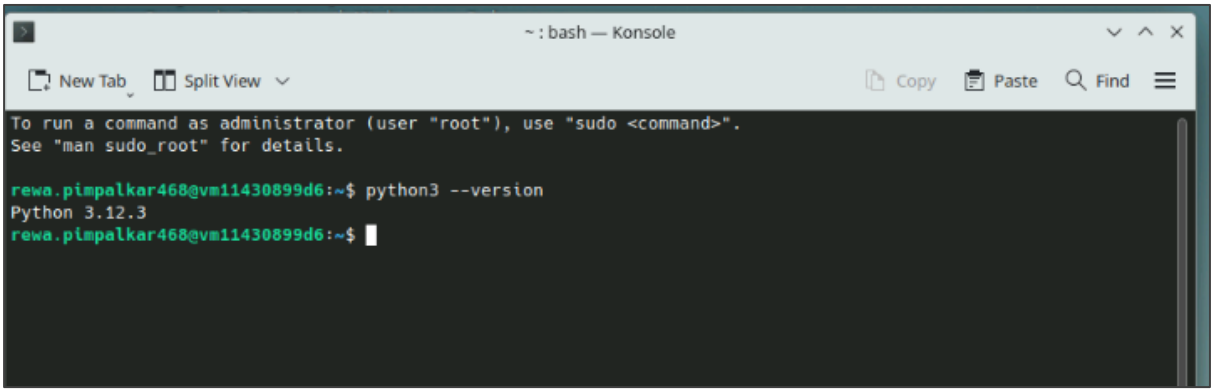
Steps to be followed:

1. Set up the format negotiation code
2. Try scenario variations in the format negotiation code

Step 1: Set up the format negotiation code

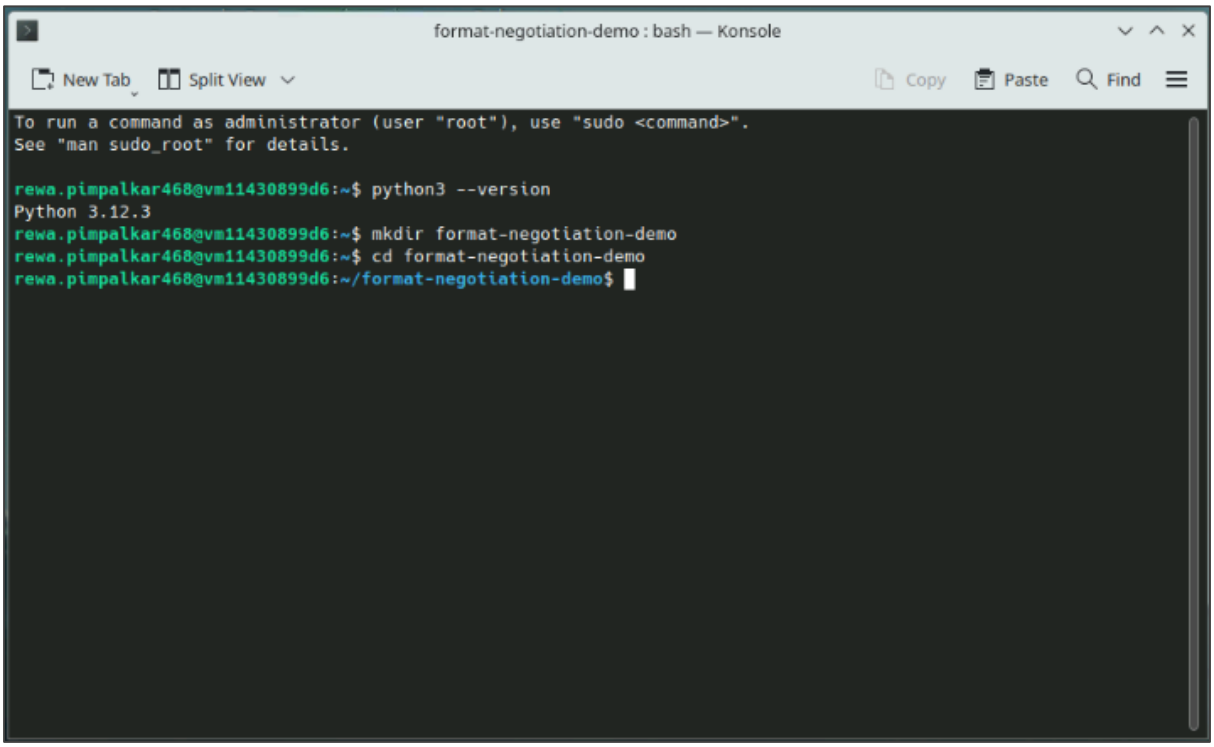
1.1 Open the command prompt and check if Python is installed using the following command:

python3 --version

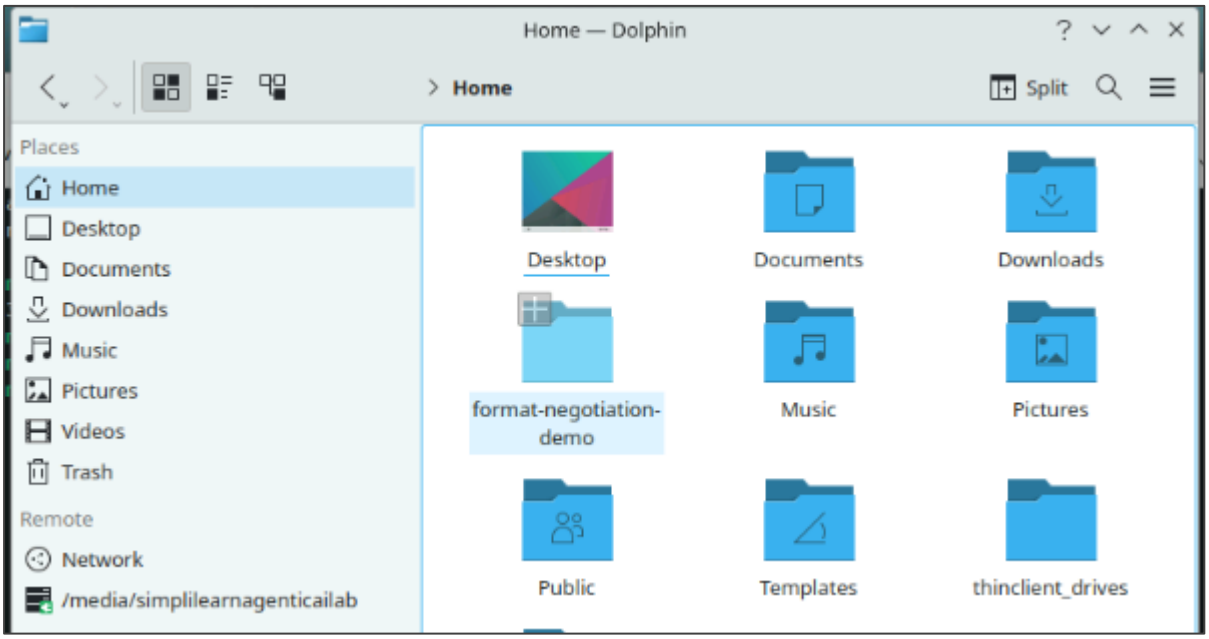


1.2 Create a new directory **format-negotiation-demo** and change into the newly created directory using the following commands:

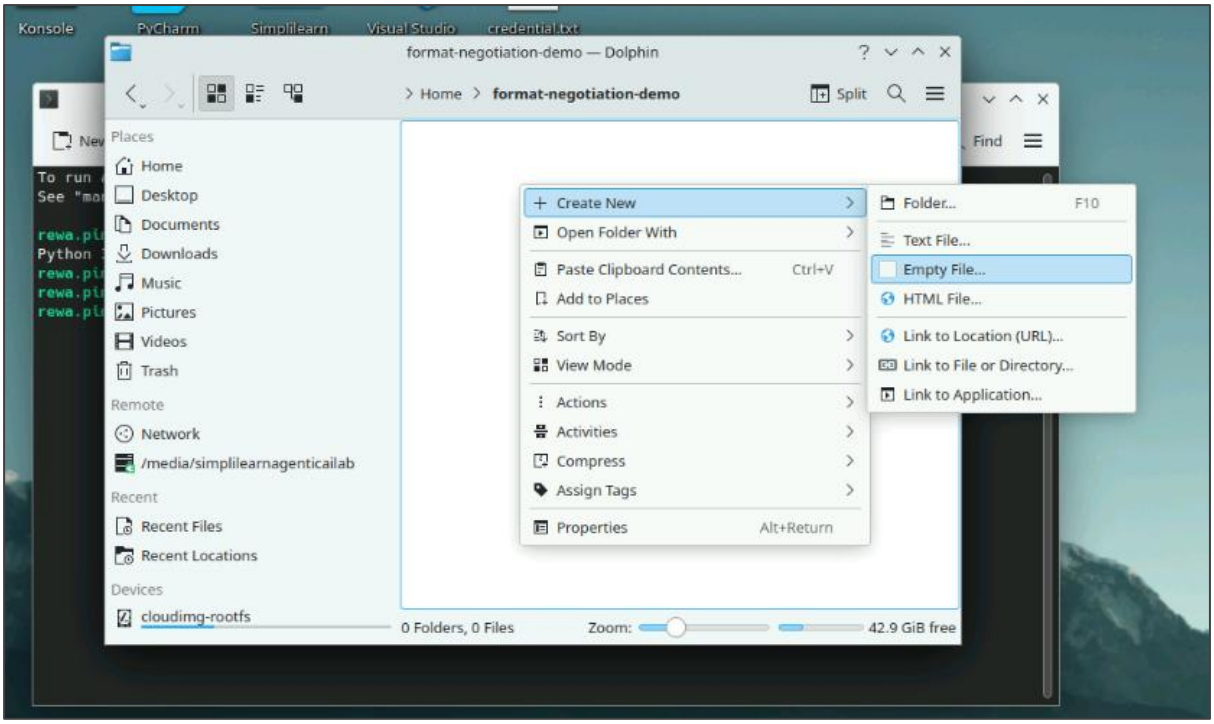
mkdir format-negotiation-demo
cd format-negotiation-demo



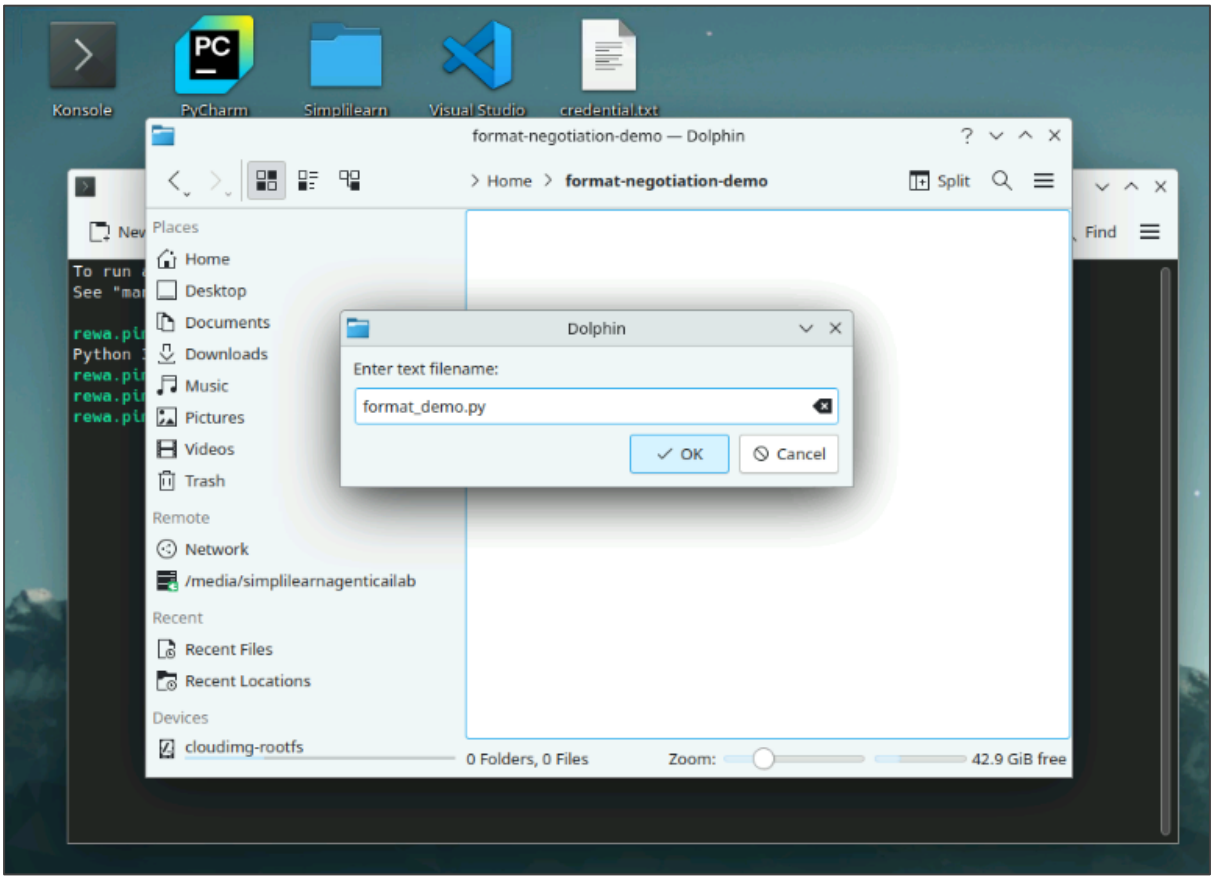
1.3 Locate and open the **format-negotiation-demo** folder



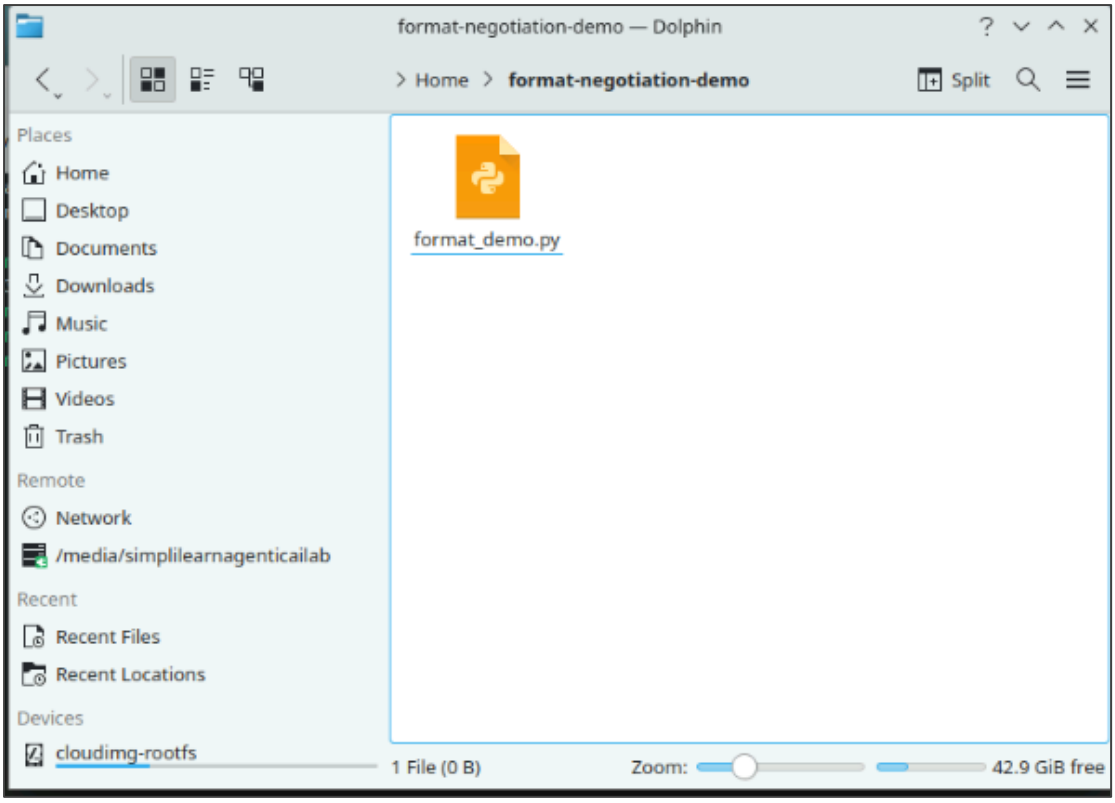
1.4 Right-click and create a new file using the option **Create New** > **Empty File**



1.5 Name the file **format_demo.py** and then click **OK**



The **format_demo.py** file is created.



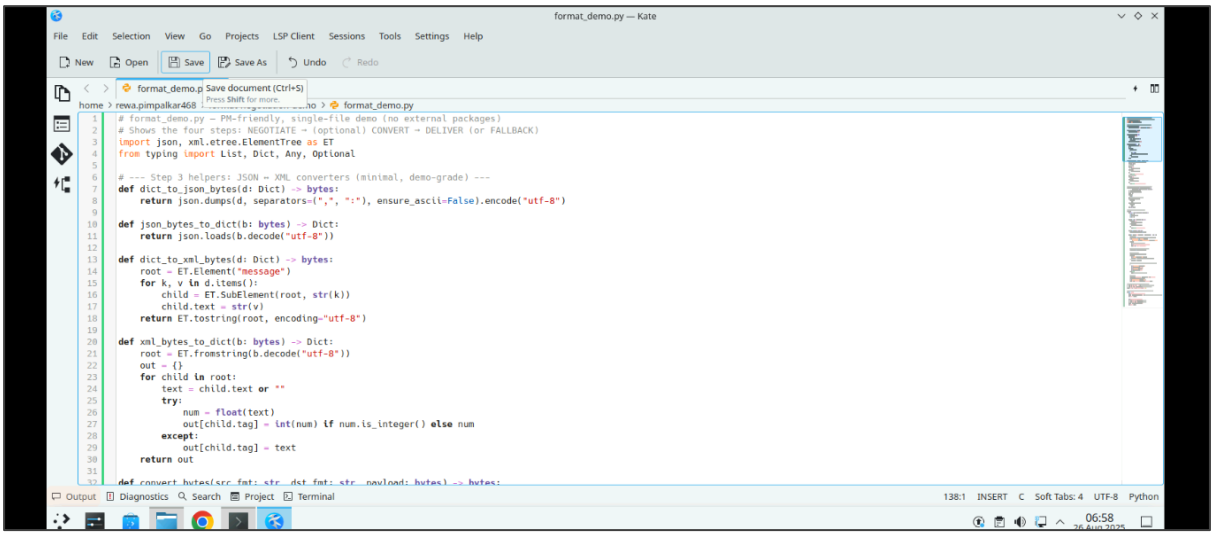
1.6 Open **format_demo.py**, paste the following code into this file, and save it:

```
# format_demo.py — PM-friendly, single-file demo (no external packages)
# Shows the four steps: NEGOTIATE → (optional) CONVERT → DELIVER (or
FALLBACK)
import json, xml.etree.ElementTree as ET
from typing import List, Dict, Any, Optional

# --- Step 3 helpers: JSON ↔ XML converters (minimal, demo-grade) ---
def dict_to_json_bytes(d: Dict) -> bytes:
    return json.dumps(d, separators=(",", ":"), ensure_ascii=False).encode("utf-8")

def json_bytes_to_dict(b: bytes) -> Dict:
    return json.loads(b.decode("utf-8"))

def dict_to_xml_bytes(d: Dict) -> bytes:
    root = ET.Element("message")
    for k, v in d.items():
        child = ET.SubElement(root, str(k))
        child.text = str(v)
    return ET.tostring(root, encoding="utf-8")

def xml_bytes_to_dict(b: bytes) -> Dict:
    root = ET.fromstring(b.decode("utf-8"))
    out = {}
    for child in root:
```

```
        text = child.text or ""
        try:
            num = float(text)
            out[child.tag] = int(num) if num.is_integer() else num
        except:
            out[child.tag] = text
    return out

def convert_bytes(src_fmt: str, dst_fmt: str, payload: bytes) -> bytes:
    src_fmt, dst_fmt = src_fmt.upper(), dst_fmt.upper()
    if src_fmt == dst_fmt:
        return payload
    # Decode to dict
    if src_fmt == "JSON":
        d = json_bytes_to_dict(payload)
    elif src_fmt == "XML":
        d = xml_bytes_to_dict(payload)
    else:
        raise ValueError("Unsupported source format")
    # Encode to destination
    if dst_fmt == "JSON":
        return dict_to_json_bytes(d)
    elif dst_fmt == "XML":
        return dict_to_xml_bytes(d)
    else:
        raise ValueError("Unsupported destination format")

# --- Step 2: Negotiation (find overlap, respect preference order) ---
def negotiate(sender_supported: List[str], receiver_supported: List[str],
              sender_pref: List[str], receiver_pref: List[str]) -> Optional[str]:
    s = {x.upper() for x in sender_supported}
    r = {x.upper() for x in receiver_supported}
    overlap = s & r
    if not overlap:
        return None
    # Try sender preference first, then receiver
    for p in [x.upper() for x in sender_pref]:
        if p in overlap:
            return p
    for p in [x.upper() for x in receiver_pref]:
        if p in overlap:
            return p
    return next(iter(overlap))

# --- Minimal Agent model (kept simple for PMs) ---
class Agent:
    def __init__(self, name: str, supported: List[str], pref: List[str]):
```

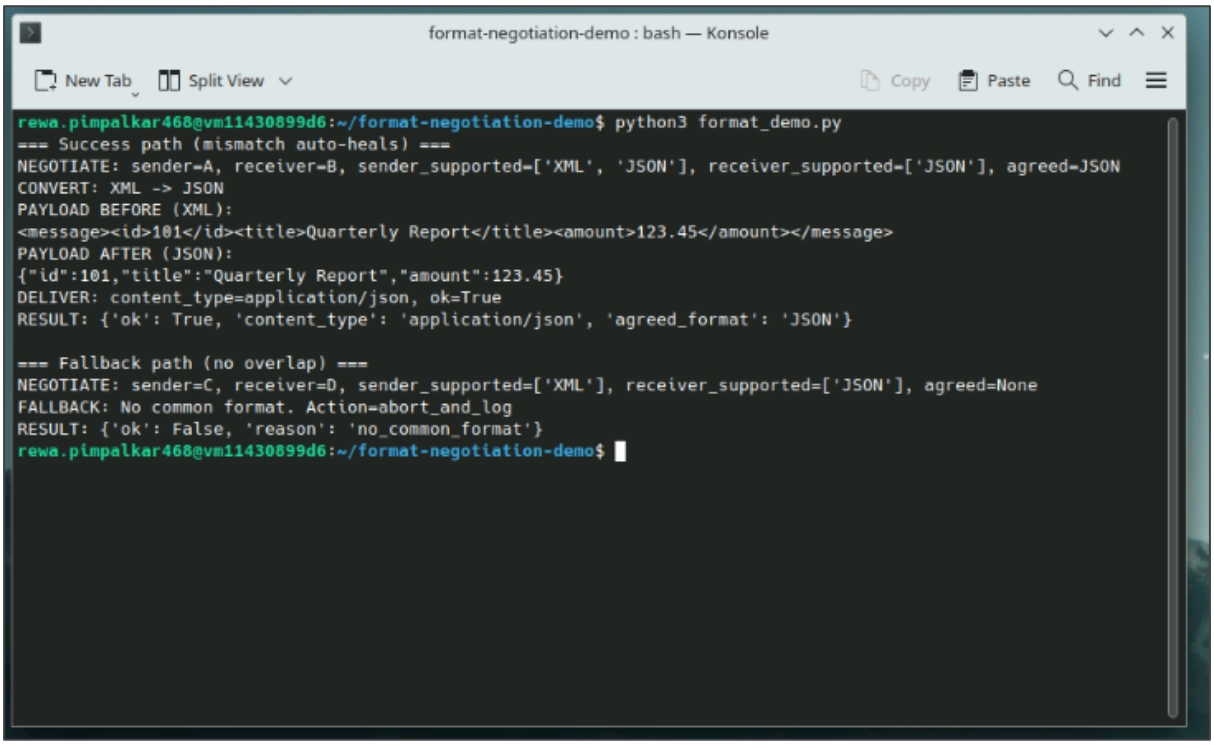
```
        self.name = name
        self.supported = supported
        self.pref = pref

    def encode(self, fmt: str, message: Dict[str, Any]) -> bytes:
        if fmt.upper() == "JSON":
            return dict_to_json_bytes(message)
        elif fmt.upper() == "XML":
            return dict_to_xml_bytes(message)
        else:
            raise ValueError("Unsupported encode format")

    def receive(self, payload: bytes, fmt: str) -> bool:
        return fmt.upper() in [x.upper() for x in self.supported]

    def send(self, receiver: "Agent", message: Dict[str, Any], src_format: str) ->
Dict[str, Any]:
        # STEP 2: NEGOTIATE
        agreed = negotiate(self.supported, receiver.supported, self.pref, receiver.pref)
        print(f"NEGOTIATE: sender={self.name}, receiver={receiver.name}, "
              f"sender_supported={self.supported},
receiver_supported={receiver.supported}, agreed={agreed}")
        if not agreed:
            # STEP 4 (fallback path when no overlap)
            print("FALLBACK: No common format. Action=abort_and_log")
            return {"ok": False, "reason": "no_common_format"}

        # Prepare "before" payload in the sender's chosen format
        payload_before = self.encode(src_format, message)

        # STEP 3: CONVERT if needed
        needs_conv = src_format.upper() != agreed.upper()
        if needs_conv:
            print(f"CONVERT: {src_format.upper()} -> {agreed.upper()}")
            payload_after = convert_bytes(src_format, agreed, payload_before)
        else:
            payload_after = payload_before

        # (Nice to show) Print payload before/after so PMs can see the wrapper
change
        try:
            print(f"PAYLOAD BEFORE ({src_format.upper()}):")
            print(payload_before.decode("utf-8"))
            print(f"PAYLOAD AFTER ({agreed.upper()}):")
            print(payload_after.decode("utf-8"))
        except Exception:
            pass  # keep demo robust even if decoding fails
```

```
        # STEP 4: DELIVER
        content_type = "application/json" if agreed.upper() == "JSON" else
"application/xml"
        ok = receiver.receive(payload_after, agreed)
        print(f"DELIVER: content_type={content_type}, ok={ok}")
        return {"ok": ok, "content_type": content_type, "agreed_format": agreed}

# --- STEP 1: Define Supported Formats per Agent (metadata) ---
A = Agent("A", ["XML", "JSON"], ["XML", "JSON"])  # Sender supports XML+JSON,
prefers XML
B = Agent("B", ["JSON"], ["JSON"])              # Receiver supports JSON only
message = {"id": 101, "title": "Quarterly Report", "amount": 123.45}

# --- STEP 4: Test Interoperability (run both scenarios when executed) ---
if __name__ == "__main__":
    print("=== Success path (mismatch auto-heals) ===")
    result1 = A.send(B, message, src_format="XML")  # A sends XML, agreed will be
JSON → convert, then deliver
    print("RESULT:", result1, "\n")

    print("=== Fallback path (no overlap) ===")
    C = Agent("C", ["XML"], ["XML"])              # only XML
    D = Agent("D", ["JSON"], ["JSON"])             # only JSON
    result2 = C.send(D, message, src_format="XML")
    print("RESULT:", result2)
```



1.7 Run the **format_demo.py** file using the following command:

python3 format_demo.py

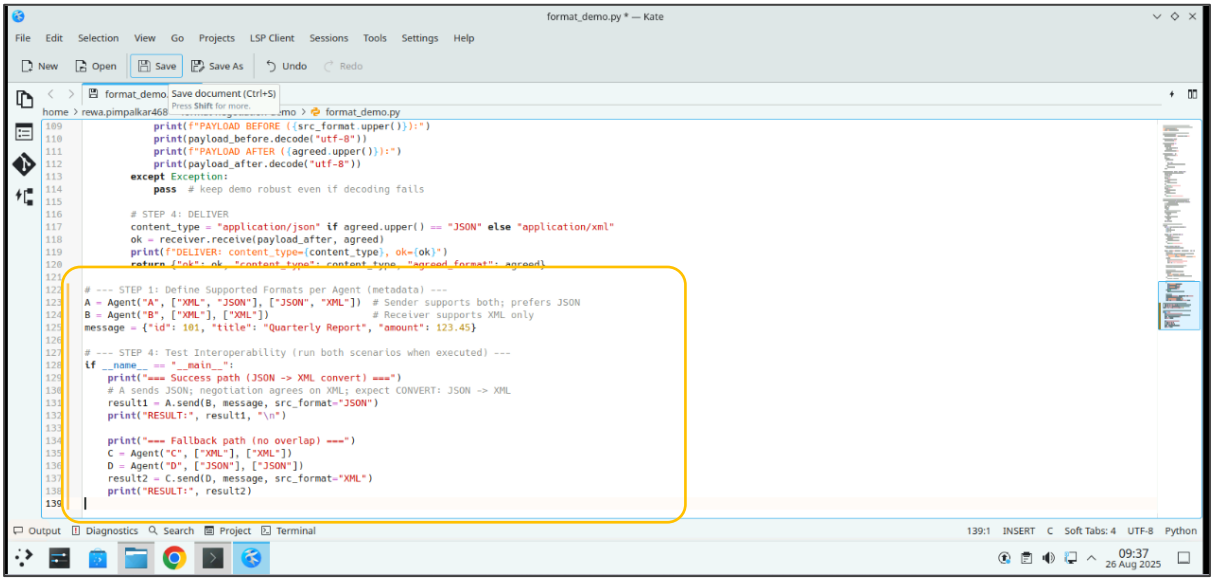


Note: This scenario negotiates JSON, auto-converts XML → JSON, and delivers successfully; the second scenario finds no shared format and triggers a safe FALLBACK.

Step 2: Try scenario variations in the format negotiation code

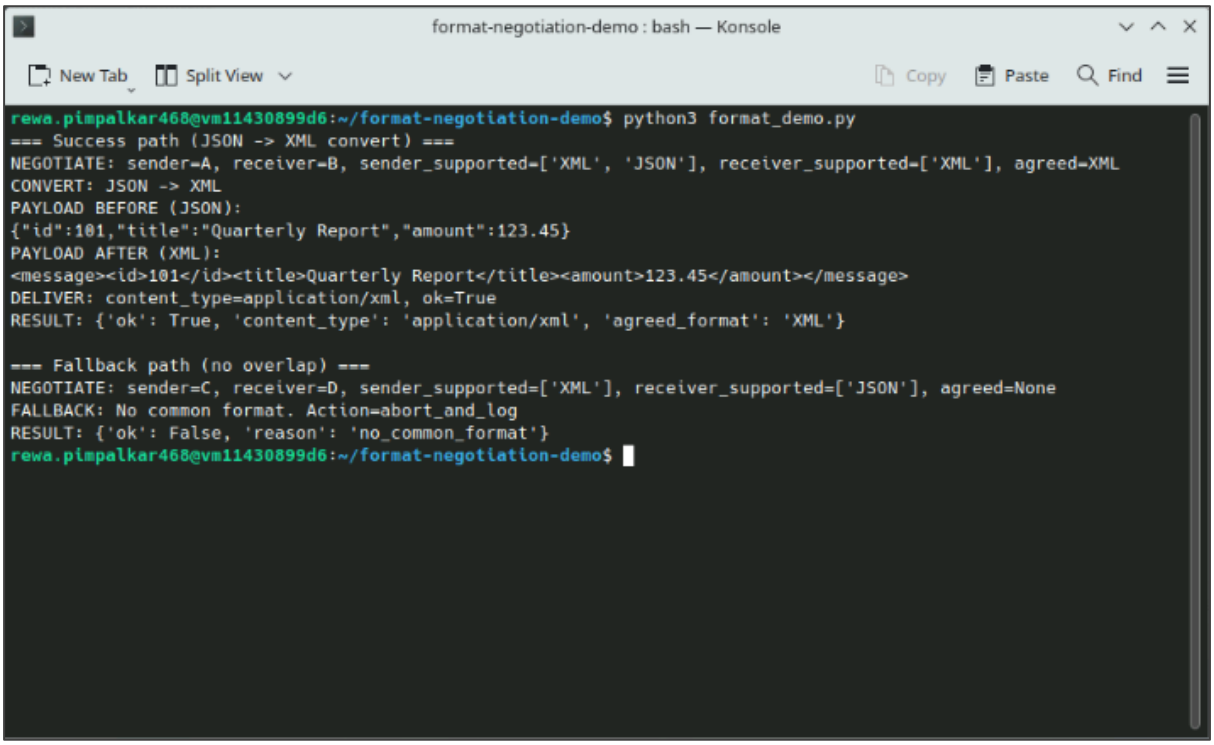
2.1 Edit the agent definitions at the bottom of **format_demo.py** with the following code to test JSON → XML conversion, then save the file:

```
# --- STEP 1: Define Supported Formats per Agent (metadata) ---
A = Agent("A", ["XML", "JSON"], ["JSON", "XML"])  # Sender supports both; prefers JSON
B = Agent("B", ["XML"], ["XML"])                  # Receiver supports XML only
message = {"id": 101, "title": "Quarterly Report", "amount": 123.45}

# --- STEP 4: Test Interoperability (run both scenarios when executed) ---
if __name__ == "__main__":
    print("=== Success path (JSON -> XML convert) ===")
    # A sends JSON; negotiation agrees on XML; expect CONVERT: JSON -> XML
    result1 = A.send(B, message, src_format="JSON")
    print("RESULT:", result1, "\n")

    print("=== Fallback path (no overlap) ===")
    C = Agent("C", ["XML"], ["XML"])
    D = Agent("D", ["JSON"], ["JSON"])
    result2 = C.send(D, message, src_format="XML")
    print("RESULT:", result2)
```



2.2 Rerun the code using the command given below:

python3 format_demo.py

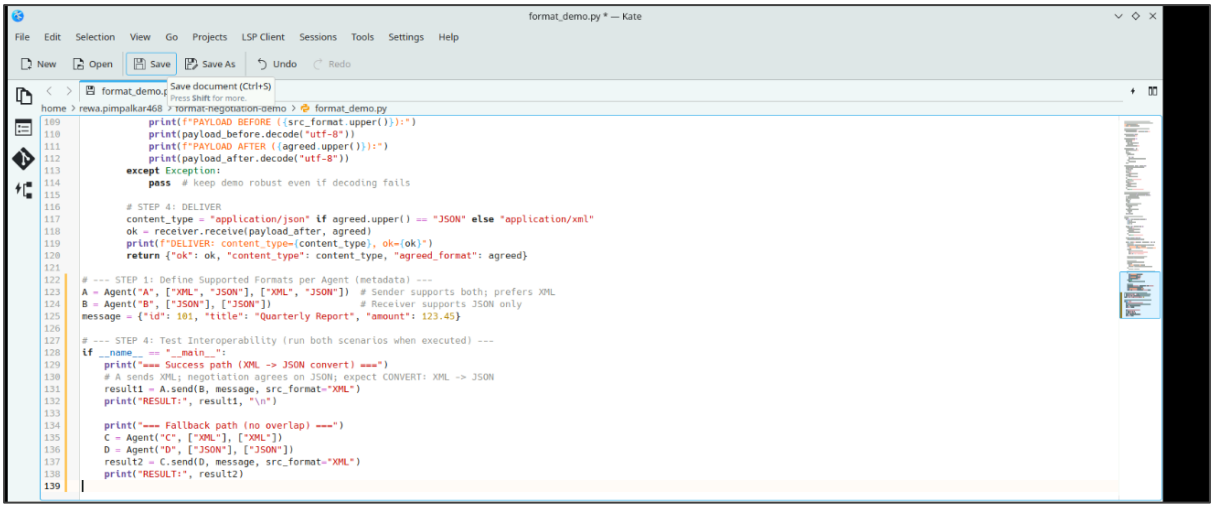


> **Note:** The Success path: agents negotiate **XML**, auto-convert the sender's **JSON → XML**, and deliver successfully; then a **no-overlap** scenario triggers a safe **FALLBACK**.

2.3 Edit the code at the bottom of **format_demo.py** with the following code to test XML → JSON conversion, then save the file:

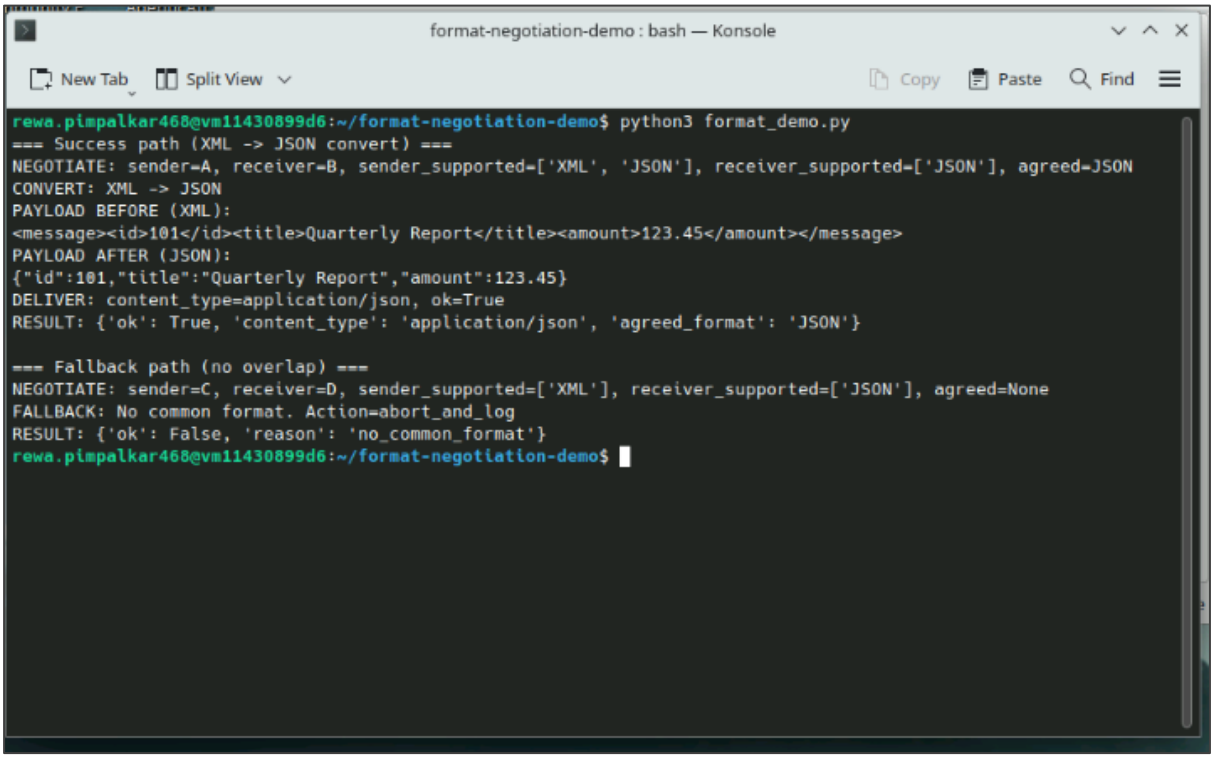
```
# --- STEP 1: Define Supported Formats per Agent (metadata) ---
A = Agent("A", ["XML", "JSON"], ["XML", "JSON"])  # Sender supports both; prefers XML
B = Agent("B", ["JSON"], ["JSON"])              # Receiver supports JSON only
message = {"id": 101, "title": "Quarterly Report", "amount": 123.45}

# --- STEP 4: Test Interoperability (run both scenarios when executed) ---
if __name__ == "__main__":
    print("=== Success path (XML -> JSON convert) ===")
    # A sends XML; negotiation agrees on JSON; expect CONVERT: XML -> JSON
    result1 = A.send(B, message, src_format="XML")
    print("RESULT:", result1, "\n")

    print("=== Fallback path (no overlap) ===")
    C = Agent("C", ["XML"], ["XML"])
    D = Agent("D", ["JSON"], ["JSON"])
    result2 = C.send(D, message, src_format="XML")
```

print("RESULT:", result2)



2.4 Rerun the code using the command given below:

python3 format_demo.py



Note: The Success case: agents agree on **JSON**, auto-convert the sender's **XML → JSON**, and deliver; then the **no-overlap** scenario aborts safely with a logged **FALLBACK**.

By following these steps, you have successfully demonstrated response-format negotiation end-to-end: agents declare capabilities, agree on a common format, auto-convert when required, and fail safely with clear logs.